

Power BI Concepts

What is Power BI?

Power BI is Microsoft's Business Intelligence (BI) platform used for data analysis, reporting, and dashboard creation. It allows users to perform the complete BI workflow—from connecting to data sources, transforming data, creating relationships, building dashboards, performing calculations, and publishing reports—all within a single ecosystem.

Power BI was developed as an evolution of Microsoft's data analytics capabilities (including Power Query in Excel) to provide a complete Business Intelligence solution.

Compared to Tableau:

- **Power BI**
 - Better integration with Microsoft products (Excel, SQL Server, Azure, Office 365)
 - Lower licensing cost
 - Strong data transformation using Power Query
 - Powerful data modeling and DAX support
 - **Tableau**
 - More flexibility for advanced visualizations
 - Often preferred for highly customized analytical dashboards
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Power BI Components

Power BI consists of three main services:

Power BI Desktop

The primary development environment where data is imported, cleaned, modeled, analyzed, and dashboards are created.

Power BI Service

Cloud platform used to publish reports, share dashboards, schedule data refreshes, and collaborate with others.

Power BI Mobile

Allows users to interact with published dashboards from mobile devices.

ETL Process in Power BI

Power BI follows an ETL (Extract → Transform → Load) workflow.

1. Extract

Data can be imported from multiple sources such as:

- Excel
 - CSV
 - SQL Databases
 - Azure
 - Web APIs
 - SharePoint
 - Google Analytics
 - Many other connectors
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2. Transform (Power Query)

If the dataset requires cleaning, it is first opened in **Power Query Editor**.

Common transformations include:

- Changing data types
- Removing duplicates
- Filtering rows
- Sorting data
- Handling null values
- Renaming columns
- Splitting or merging columns
- Replacing values
- Creating custom columns
- Grouping rows
- Appending tables
- Merging queries

An important feature of Power Query is that every transformation is automatically recorded as an **Applied Step**, making the entire cleaning process reproducible and easy to modify.

After all transformations are complete, selecting **Close & Apply** loads the cleaned data into Power BI.

3. Load

The transformed data is loaded into Power BI's data model, where relationships, calculations, and visualizations are created.

Power BI Views

Report View

The Report View is used to build dashboards using different visualizations such as:

- KPI Cards
- Line Charts
- Bar Charts
- Column Charts
- Pie Charts
- Donut Charts
- Maps
- Tables
- Matrix
- Treemaps
- Slicers

This is where the final interactive dashboard is designed.

Data View

Displays the underlying rows of each table.

Useful for:

- Inspecting imported data
 - Creating calculated columns
 - Creating calculated tables
 - Writing DAX expressions
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Model View

Used for **data modeling**.

It displays relationships between tables similar to an ER diagram.

Relationships can be created by dragging matching columns between tables.

Important concepts:

- One-to-One
- One-to-Many
- Many-to-Many
- Cross Filter Direction
- Active vs Inactive Relationships

Good data modeling improves report performance and accuracy.

DAX (Data Analysis Expressions)

DAX is Power BI's formula language used for calculations.

It is used to create:

- Calculated Columns
- Measures
- Calculated Tables

Examples of common DAX functions:

- COALESCE()
- IF()
- SUM()
- SUMX()
- AVERAGE()
- COUNT()
- DISTINCTCOUNT()
- CALCULATE()
- FILTER()
- RELATED()
- DIVIDE()

In this project, I used:

`Cleaned_Score = COALESCE(CustomerScore,0)`

to replace NULL customer scores.

Difference to remember:

- **Calculated Column** → Computed for every row and stored in the model.
 - **Measure** → Calculated dynamically based on filters and visuals.
Preferred for aggregations and KPIs because they are more memory efficient.
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Slicers

Slicers are interactive filters that allow users to filter every visualization on the page.

Examples:

- Country
- Date
- Category
- Customer

They make dashboards dynamic and user-friendly.

Drill Down

Drill Down enables users to navigate through different levels of data hierarchy.

Example:

Year → Quarter → Month → Day

This allows users to analyze trends at different levels of detail.

KPIs

KPI cards summarize important business metrics such as:

- Total Revenue
- Total Sales
- Profit
- Number of Orders
- Average Customer Rating

They provide an at-a-glance overview of business performance.

Data Modeling Best Practices

Some important practices include:

- Use a Star Schema whenever possible.
- Keep dimension tables separate from fact tables.
- Avoid duplicate data.

- Use Measures instead of Calculated Columns whenever possible.
 - Hide unnecessary columns.
 - Use meaningful table and column names.
 - Create relationships carefully to avoid ambiguity.
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Workflow & Key Learnings

Step 1 — Import Data

Imported the dataset into Power BI Desktop.

Step 2 — Clean Data (Power Query)

Performed preprocessing using Power Query:

- Merged `First_Name` and `Last_Name` into a new `Customer_Name` column.
 - Converted `Order_Date` from Text to Date datatype.
 - Reviewed and verified data types.
 - Applied all transformations using **Close & Apply**.
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Step 3 — Build Data Model

Verified relationships between tables in Model View.

If relationships were missing:

- Connected tables using common columns.
 - Selected the appropriate cardinality.
 - Ensured relationships matched the business logic.
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Step 4 — Create DAX Calculations

Created a calculated column:

```
Cleaned_Score = COALESCE(CustomerScore,0)
```

to replace missing customer scores.

Step 5 — Build Dashboard

Designed an interactive Sales Dashboard containing:

- KPI Cards
- Revenue Trend Line Chart
- Product-wise Sales Breakdown
- Country-wise Sales Distribution
- Category-wise Sales Analysis
- Pie Charts
- Donut Charts
- Bar Charts

Step 6 — Add Interactivity

Added slicers to allow users to filter the dashboard by:

- Country
- Date

Implemented Drill Down functionality for the revenue trend:

Year → Quarter → Month

Step 7 — Dashboard Formatting

Improved dashboard usability by:

- Resizing visuals
- Aligning components
- Applying consistent formatting
- Editing titles and labels
- Adjusting colors and themes where necessary

Most formatting options can be accessed by selecting a visual, after which the Visualizations pane and ribbon expose customization options.

Key Learnings

- Power BI follows an **ETL workflow** where data is extracted, transformed using Power Query, and loaded into the data model.
- **Power Query** is intended for data cleaning and preprocessing before analysis.
- **DAX** is used for analytical calculations after the data has been loaded.
- **Calculated Columns** operate row by row, while **Measures** are evaluated dynamically based on the report context.
- The **Model View** defines relationships between tables and is essential for accurate analysis.
- **Slicers** and **Drill Down** make dashboards interactive and improve user experience.
- Well-structured data models and clean transformations make reports easier to maintain and more performant.
- Building dashboards is not just about creating charts—it involves understanding the business problem, choosing the right KPIs, modeling the data correctly, and presenting insights in a clear, actionable way.